

# Patrick Tierney

(319) 330-5881 | [ptierney13@gmail.com](mailto:ptierney13@gmail.com) | Kirkland, WA 98033 | [linkedin.com/in/ptierney13](https://www.linkedin.com/in/ptierney13)

## EXPERIENCE

---

### Microsoft

**Software Engineer II** (2019 – 2025), **Software Engineer I** (2017 – 2019)

#### CI/CD & Internal Tooling

- Architected and led delivery of a testing platform used by 150+ engineers supporting Copilot Hub, standardizing automated testing
- Defined standards and requirements for an AI-driven static analysis system enforcing testing and compliance
- Built and managed CI/CD for Whiteboard and Copilot Hub, automating builds, testing, and validation
- Owned the Azure DevOps Artifacts npm feed for Copilot Hub shared components, managing package publishing, PR-based promotion, rollback, and CI-validated updates
- Consolidated disparate AI tooling from across Microsoft teams into a unified, maintainable platform codebase

#### Observability & Telemetry

- Designed shared telemetry APIs, PR-time static analysis, and ingestion-time checks that blocked sensitive data from being logged
- Built environment-aware logging APIs used across Whiteboard so engineers could use one telemetry interface while keeping data from sovereign government cloud environments (GCC, GCCH, DoD) out of non-GCC systems
- Enhanced a telemetry query DSL used by 1,000+ Edge engineers, improving analytics and diagnostic capabilities
- Developed automation that generated dashboards from telemetry data for new feature rollouts

#### Reliability & Incident Response

- Established on-call and incident response practices, reducing incident resolution time by 80% and engineering support workload by 70%
- Maintained 99.9% availability for Microsoft Whiteboard supporting school systems globally during COVID-19
- Built tooling and operational runbooks enabling customer support to resolve issues without engineer escalation

#### Backend Services & APIs

- Developed Azure microservices enabling secure credential and certificate management for internal infrastructure
- Implemented OAuth-based authentication and JWT access token validation for Whiteboard backend services, integrating with Azure AD scopes and permissions

#### Platform & Framework Development

- Drove migration of Whiteboard's layout engine from C# to TypeScript/React within its backing framework
- Built layout-engine tooling to position visual elements correctly on screen, supporting 80+ engineers as Whiteboard scaled to tens of millions of users
- Owned layout logic for on-screen placement for non-hierarchical graphical elements across 50+ internal applications

## TECHNICAL SKILLS

---

**Languages & Frameworks:** TypeScript, JavaScript, C#, .NET, SQL, React, Python, C++, R

**Infrastructure & Automation:** Azure services, Azure DevOps, YAML-based CI/CD, PowerShell

**Testing, Observability & Security:** Jest, Playwright, OpenTelemetry, Power BI, OAuth/JWT authentication, Distributed Tracing & Observability

## EDUCATION

---

**Harvey Mudd College**

Bachelor's Degree, Computer Science and Mathematics; Concentration in Economics

Claremont, CA

May 2016